

A NON-UNIFORMLY SMOOTH SPACE WITH PROPERTY $\mathcal{D}(s, c_1)$

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STATUS

This is a candidate full answer to Remark 2.11 of [1]. The answer is positive: property $\mathcal{D}(s, c_1)$ can hold for a non-uniformly smooth Banach space and a dictionary.

SOURCE QUESTION

Definition 1.12 of [1] says that $(X, \|\cdot\|, \mathcal{D})$ has property $\mathcal{D}(s, c_1)$ if, for some $s > 1$ and $c_1 > 0$,

$$\text{dist}(f, [\varphi]) \leq \|f\| (1 - c_1 |F_f(\varphi)|^s)$$

for every $f \neq 0$ and every $\varphi \in \mathcal{D}$, where F_f is an associated norming functional. Remark 2.11 asks whether this property could hold for some non-uniformly smooth Banach space and some dictionary.

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COROLLARY 2.9. *Suppose that $\rho(t) \leq \gamma t^q$, for some $\gamma > 0$ and $q > 1$. Then, for every dictionary $\mathcal{D}$, the pair $(\mathbb{X}, \mathcal{D})$ has the property $\mathcal{D}(s, c_1)$ with $s = q'$ and some $c_1 = c_1(\gamma, q) > 0$.*

REMARK 2.10. Observe that $\rho(t) \leq \gamma t^q$ can only hold for $q \leq 2$, and hence $q' \geq 2$. However, property $\mathcal{D}(s, c_1)$ may hold in some cases with s close to 1. For instance, if $\mathbb{X} = \ell^p$ with $p > 2$ and $\mathcal{D}$ is the canonical basis, then it is not hard to verify that $(\mathbb{X}, \mathcal{D})$ has property $\mathcal{D}(s, c_1)$ with $s = p'$ and $c_1 = 1/p$; see Proposition 4.12 below for a more general class of such examples.

REMARK 2.11. Observe that the property $\mathcal{D}(s, c_1)$ can also be written as

$$|F_f(\varphi)| \leq c_1^{-1} \left(1 - \text{dist}(f, [\varphi])\right)^{1/s}, \quad \forall \|f\| = 1, \varphi \in \mathcal{D}.$$

One may ask whether this property could hold for some non-uniformly smooth Banach space $(\mathbb{X}, \|\cdot\|)$ and some dictionary $\mathcal{D}$.

PROOF INTUITION

The example is a Hilbert norm with a small absolute-value kink:

$$N_\varepsilon(x, y) = (x^2 + y^2)^{1/2} + \varepsilon|x + y|.$$

The kink makes the norm non-smooth, hence not uniformly smooth. But the kink is transverse to the two coordinate dictionary lines. For each coordinate line, the best distance in the N_ε -norm is obtained by a one-variable convex minimization. The distance defect from this minimum is quadratic in the chosen norming functional's value on the coordinate atom. This gives $\mathcal{D}(2, c)$ on the two-dimensional block. Taking an ℓ_2 -sum with a Hilbert space preserves this coordinate-dictionary estimate and gives an infinite-dimensional example.

THE EXAMPLE

Fix $0 < \varepsilon < 1/2$. On $\mathbb{R}^2$, define

$$N_\varepsilon(x, y) = \sqrt{x^2 + y^2} + \varepsilon|x + y|.$$

Let $E_\varepsilon = (\mathbb{R}^2, N_\varepsilon)$, and let $H = \ell_2$. Set

$$X = E_\varepsilon \oplus_2 H, \quad \|(z, h)\|_X = (N_\varepsilon(z)^2 + \|h\|_2^2)^{1/2}.$$

Let

$$u_1 = \frac{(1, 0)}{1 + \varepsilon}, \quad u_2 = \frac{(0, 1)}{1 + \varepsilon},$$

which are unit vectors in E_ε , and let (e_n) be the usual unit vector basis of H . The dictionary is

$$\mathcal{D} = \{(u_1, 0), (u_2, 0)\} \cup \{(0, e_n) : n \geq 1\}.$$

Its linear span is dense in X .

FINITE-DIMENSIONAL BLOCK ESTIMATE

For $z = (x, y) \neq 0$, put $r = (x^2 + y^2)^{1/2}$. Choose the following norming functional for E_ε :

$$G_z(a, b) = \frac{xa + yb}{r} + \varepsilon\theta_z(a + b),$$

where $\theta_z = \text{sign}(x + y)$ if $x + y \neq 0$, and $\theta_z = 0$ if $x + y = 0$. This is a subgradient of N_ε at z , hence $G_z(z) = N_\varepsilon(z)$ and $\|G_z\|_{E_\varepsilon^*} = 1$.

Lemma 1. *There is a constant $c_E = c_E(\varepsilon) > 0$ such that, for $j = 1, 2$,*

$$\text{dist}_{E_\varepsilon}(z, [u_j]) \leq N_\varepsilon(z)(1 - c_E|G_z(u_j)|^2), \quad z \neq 0.$$

Proof. We give the proof for $j = 1$; the other case is symmetric. Since $[u_1] = [(1, 0)]$, the distance from $z = (x, y)$ to this line is

$$\delta_1(z) = \inf_{\lambda \in \mathbb{R}} N_\varepsilon(x - \lambda, y) = \inf_{t \in \mathbb{R}} \left(\sqrt{t^2 + y^2} + \varepsilon|t + y| \right).$$

If $y = 0$, then $\delta_1(z) = 0$ and the desired inequality is immediate after decreasing $c_E \leq 1$. Assume $y \neq 0$, write $\eta = \text{sign } y$, and set $t = x/|y|$. By homogeneity,

$$\delta_1(z) = |y|m_\eta, \quad N_\varepsilon(z) = |y|h_\eta(t),$$

where

$$h_\eta(t) = \sqrt{t^2 + 1 + \varepsilon|t + \eta|}, \quad m_\eta = \inf_{v \in \mathbb{R}} h_\eta(v).$$

For $0 < \varepsilon < 1/2$, the unique minimizer of h_η is

$$t_\eta = -\eta \frac{\varepsilon}{\sqrt{1 - \varepsilon^2}},$$

which is not the kink point $-\eta$. Away from the kink, the selected value $(1 + \varepsilon)G_z(u_1)$ is exactly the derivative $h'_\eta(t)$. At the kink our selected value is one element of the subdifferential, while $h_\eta(-\eta) > m_\eta$.

Consider, where the denominator is nonzero,

$$Q_\eta(t) = \frac{h_\eta(t) - m_\eta}{h_\eta(t) |h'_\eta(t)/(1 + \varepsilon)|^2},$$

using the selected subgradient value at the kink. This quotient has a positive lower bound. Indeed, outside a compact interval it has positive liminf; away from t_η and the kink this is ordinary continuity; at the kink the numerator is strictly positive; and near t_η ,

$$h_\eta(t) - m_\eta = \frac{1}{2}h''_\eta(t_\eta)(t - t_\eta)^2 + o((t - t_\eta)^2), \quad h'_\eta(t) = h''_\eta(t_\eta)(t - t_\eta) + o(t - t_\eta),$$

with $h''_\eta(t_\eta) > 0$. Hence Q_η extends there with a positive finite limit. Taking the minimum over $\eta = \pm 1$ gives $c_E > 0$. Rewriting the inequality $Q_\eta(t) \geq c_E$ in homogeneous variables proves the claim for u_1 . $\square$

HILBERT SUM STABILITY

The Hilbert basis has the corresponding estimate with exponent 2: if $a_n = \langle h, e_n \rangle / \|h\|_2$, then

$$\text{dist}_H(h, [e_n]) = \|h\|_2(1 - |a_n|^2)^{1/2} \leq \|h\|_2(1 - \frac{1}{2}|a_n|^2).$$

Choose norming functionals on X as follows. If $z \neq 0$ and $h \neq 0$, with $R = \|(z, h)\|_X$, set

$$F_{(z, h)}(w, k) = \frac{N_\varepsilon(z)}{R} G_z(w) + \frac{\|h\|_2}{R} \left\langle \frac{h}{\|h\|_2}, k \right\rangle.$$

The evident one-component formula is used when $z = 0$ or $h = 0$.

Theorem 1. *The Banach space $X = E_\varepsilon \oplus_2 \ell_2$ is not uniformly smooth, but $(X, \mathcal{D})$ has property $\mathcal{D}(2, c)$ for some $c = c(\varepsilon) > 0$.*

Proof. The norm N_ε is not Gâteaux differentiable at any nonzero point of the line $x + y = 0$, because the term $|x + y|$ has a kink there. Therefore the norm of X is not Gâteaux differentiable at $((1, -1), 0)$. Uniform smoothness implies Gâteaux differentiability away from zero, so X is not uniformly smooth.

It remains to prove property $\mathcal{D}(2, c)$. Let $R = \|(z, h)\|_X$, $A = N_\varepsilon(z)$, and $B = \|h\|_2$. First take an atom $\varphi = (u_j, 0)$. Put $a = G_z(u_j)$, with $a = 0$ if $z = 0$. By Lemma 1,

$$\text{dist}_{E_\varepsilon}(z, [u_j]) \leq A(1 - c_E|a|^2).$$

Hence

$$\begin{aligned} \text{dist}_X((z, h), [\varphi])^2 &\leq A^2(1 - c_E|a|^2)^2 + B^2 \\ &\leq R^2 \left(1 - c_E \frac{A^2}{R^2}|a|^2\right). \end{aligned}$$

Since $|F_{(z, h)}(\varphi)| = (A/R)|a|$, taking square roots yields

$$\text{dist}_X((z, h), [\varphi]) \leq R \left(1 - \frac{c_E}{2}|F_{(z, h)}(\varphi)|^2\right).$$

For a Hilbert atom $\varphi = (0, e_n)$, the same computation, using the Hilbert estimate above in place of Lemma 1, gives

$$\text{dist}_X((z, h), [\varphi]) \leq R \left(1 - \frac{1}{4}|F_{(z, h)}(\varphi)|^2\right).$$

Thus property $\mathcal{D}(2, c)$ holds with

$$c = \min\{c_E/2, 1/4\} > 0.$$

□

HISTORY: WHY THE FIRST OBSTRUCTION WAS NOT THE END

Before finding the positive example above, the direct attack produced a useful negative result for a natural class of nonsmooth spaces. It is included here so the full packet records the route cleanly.

Theorem 2 (Historical obstruction). *Let E and F be nonzero Banach spaces, and let*

$$Y = E \oplus_1 F, \quad \|(e, f)\| = \|e\| + \|f\|.$$

Then no nonempty dictionary in Y can satisfy property $\mathcal{D}(s, c)$ for any $s > 1$ and $c > 0$.

Proof. It is enough to show that every unit atom $\varphi = (u, v) \in Y$ fails the one-atom inequality. Put $A = \|u\|$, $B = \|v\|$, so $A + B = 1$.

Assume first $A, B > 0$, and write $e = u/A$, $y = v/B$. If $A \leq B$, take

$$x_t = (e, ty), \quad t > 0.$$

Then $\|x_t\| = 1 + t$. Every norming functional for x_t evaluates φ as 1. On the other hand, for any scalar λ ,

$$\|x_t - \lambda\varphi\| = |1 - \lambda A| + |t - \lambda B| \geq |1 - \lambda A| + \frac{A}{B}|t - \lambda B| \geq 1 - \frac{A}{B}t,$$

and equality is attained at $\lambda = t/B$. Hence

$$\frac{\text{dist}(x_t, [\varphi])}{\|x_t\|} = \frac{1 - (A/B)t}{1 + t} \longrightarrow 1, \quad |F_{x_t}(\varphi)| = 1.$$

This contradicts any estimate $\text{dist}(x, [\varphi]) \leq \|x\|(1 - c|F_x(\varphi)|^s)$ with $c > 0$. If $A > B$, the same proof uses $x_t = (te, y)$.

If $A = 0$, choose any unit $e \in E$ and take $x_t = (e, tv)$. Then $\text{dist}(x_t, [\varphi]) = 1$, $\|x_t\| = 1 + t$, and every norming functional again evaluates φ as 1. Letting $t \downarrow 0$ gives the same contradiction. The case $B = 0$ is symmetric. $\square$

This obstruction explains why the successful example uses a small Hilbertian kink rather than an ℓ_1 -sum corner. The full packet's history folder contains the original partial packet and its superseded ledger record.

VERIFICATION AND NOVELTY NOTES

The proof above is independent of numerical experimentation. As a sanity check, the script `code/sample_block_ratio.py` samples the quotient in Lemma 1 for $\varepsilon = 1/4$; it found a positive sampled minimum approximately 0.23638.

Local run indexes were searched for arXiv:1909.13536, the exact title, $\mathcal{D}(s, c_1)$, non-uniformly smooth spaces, and related WCGA terms. Web searches for the exact Remark 2.11 phrasing and property $\mathcal{D}(s, c_1)$ found the source arXiv page [1] but no later paper explicitly answering the question. Novelty confidence is therefore reasonable but still subject to human literature review. The previous ℓ_1 -sum obstruction is no longer indexed as a separate partial result; it has been packed into this full result as historical context.

REFERENCES

- [1] S. Dilworth, G. Garrigós, E. Hernández, D. Kutzarova, and V. Temlyakov, *Lebesgue-type inequalities in greedy approximation*, arXiv:1909.13536.